



Department of Information Technology
Academic Year 2025 - 2026 (Even Semester)

Degree, Semester & Branch: IV Semester B.Tech. IT

Course Code & Title: CS3452 & Theory of Computation

Name of the Faculty member (s): Dr.V. Anusuya, ASCP/IT

Innovative Practice Description

- **Unit / Topic:** Unit IV / TM as Computer of Integer functions
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO14
- **Activity Chosen:** Learning by doing
- **Justification:**

In unit IV, By designing and simulating Turing Machines, students clearly understand how input is processed step-by-step to produce output, strengthening their grasp of state transitions and tape operations. The activity encourages students to think logically and systematically while constructing Turing Machines for different integer functions.

- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
 - The topic “**Turing Machine as a Computer of Integer Functions**” was assigned to Ms. Dhanya Sankari two days prior to the presentation.
 - she prepared thoroughly and handled the class on 24.03.2026.
 - The topic was presented for approximately 10 minutes. She initially explained the concept of Turing Machines and their role in computing integer functions, followed by illustrative examples such as basic arithmetic operations.
 - The implementation of the Learning by Teaching activity is shown in Figure 1.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO4	2	3	3	2	2	1	2	2	2	2	2

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO4	2	3	3	2	2	1	2	2	2	2	2
Justification for correlation	Apply basic Knowledge and fundamentals of Theory of Computation for TM as Computer of Integer functions	The students will be able to identify the need for analysis of complex engineering problems	Able to design the process flow to TM as Computer of Integer functions	Able to analyze the data to prove TM as Computer of Integer functions	The students will be able to follow ethics during the proof	The students able to function as a team TM as Computer of Integer functions	The students will be able to communicate one another and can able to present	The students able to learn and implement the concept in Natural Language Processing	The students able to apply mathematical and computing skills.	The students can Exhibit knowledge of languages in the field of Cloud Computing to develop IT solutions	The students able to exhibit knowledge in Languages to develop a quality product.

- **Images / Screenshot of the practice:**



Figure 1: Learning by Teaching Handled by Dhanya Sankari

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- Students showed active interest in understanding how a Turing Machine can be used to compute integer functions such as addition and multiplication.
 - Many students were able to follow the step-by-step execution of Turing Machines and appreciated the practical demonstration of function computation.
 - Learners recognized the importance of representing mathematical functions using computational models.
 - Stakeholders observed that the activity improved student engagement and helped in visualizing abstract concepts through hands-on practice.

- ❖ **Benefit of the practice:**

- Students developed the ability to **design and simulate Turing Machines** for basic integer functions.
 - The activity strengthened **algorithmic thinking and logical reasoning**, as students decomposed functions into machine-level operations.
 - Learners gained a clear understanding of how computation is performed at a fundamental level.
 - It enhanced students' ability to connect **theoretical concepts with practical execution**, improving retention and conceptual clarity.

- ❖ **Challenges faced in implementation:**

- Some students found it difficult to understand the **state transitions and tape movements**, especially for complex functions like multiplication.
 - The process of simulation was **time-consuming**, leading to reduced pace for some learners.

- **References:**

- ❖ https://www.ritrjpm.ac.in/images/B.Tech-IT/2025-2026/SS_IT3501_LEARNING%20BY%20TEACHING_U3.pdf.
 - ❖ https://www.ritrjpm.ac.in/images/B.Tech-IT/2024-2025/VA_CS3452_Learningbyteaching.pdf

Signature of Faculty Member

HOD